

Cash Conversion Cushion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Conversion Cushion (CCC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CCC achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 37 (27) bps/month with a t-statistic of 3.56 (2.68), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price) is 37 bps/month with a t-statistic of 3.88.

1 Introduction

Financial markets efficiently incorporate information about firms’ production characteristics into asset prices, yet significant cross-sectional variation in expected returns remains unexplained by standard risk factors. We identify a novel predictor of stock returns based on firms’ cash conversion efficiency relative to their earnings quality—the Cash Conversion Cushion (CCC). This measure captures the extent to which firms can convert accounting earnings into operating cash flows, providing insight into their operational efficiency and exposure to production-based risks. Despite extensive research on profitability and investment factors in asset pricing [Novy-Marx \(2013\)](#); [Fama and French \(2015\)](#), the role of cash conversion efficiency as a distinct source of systematic risk remains underexplored.

In a production-based asset pricing framework, firms’ cash conversion efficiency directly maps to their marginal costs of production and adjustment frictions [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). Firms with higher Cash Conversion Cushion demonstrate superior ability to transform earnings into cash flows, indicating lower operational frictions and more efficient working capital management. This efficiency translates into lower marginal costs of production, enabling these firms to better respond to productivity shocks and investment opportunities [Liu et al. \(2009\)](#). The production-based model predicts that firms with higher CCC face lower adjustment costs when scaling production, leading to higher conditional risk premia as investors demand compensation for bearing systematic production risk.

The Cash Conversion Cushion proxies for firms’ exposure to aggregate productivity shocks through its connection to investment frictions and capital allocation efficiency [Berk et al. \(1999\)](#). Firms with robust cash conversion capabilities maintain operational flexibility during economic downturns, reducing their sensitivity to financing frictions that amplify business cycle fluctuations. This operational hedging capacity affects the firm’s stochastic discount factor through its impact on the

marginal utility of production [Gomes et al. \(2003\)](#). Moreover, efficient cash conversion reduces the wedge between internal and external financing costs, allowing firms to pursue positive NPV projects without facing prohibitive adjustment costs.

Cross-sectional variation in Cash Conversion Cushion reflects heterogeneity in firms' production technologies and their exposure to systematic investment risks [Cooper and Priestley \(2011\)](#). The measure captures time-varying risk premia associated with production efficiency, as firms with higher CCC exhibit different sensitivities to aggregate investment shocks. This heterogeneity in production-based risk exposure generates predictable patterns in expected returns, consistent with rational pricing in a production economy where marginal costs and investment frictions determine equilibrium asset prices [Lin and Zhang \(2013\)](#).

We find that a value-weighted long/short trading strategy based on Cash Conversion Cushion achieves an annualized gross Sharpe ratio of 0.41, with a monthly average excess return of 34 basis points (t -statistic = 3.10). The strategy delivers a monthly alpha of 37 basis points relative to the Fama-French six-factor model, with a t -statistic of 3.56, demonstrating robust performance after controlling for market, size, value, profitability, investment, and momentum factors. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CCC strategy generates an average return of 28 basis points per month (t -statistic = 2.12), confirming that the anomaly persists in the most liquid segment of the market where production-based risks are most precisely priced.

The economic significance of the Cash Conversion Cushion effect remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.38 places the strategy in the top 6% of all documented anomalies, while the net alpha relative to the six-factor model equals 27 basis points per month (t -statistic = 2.68). Robustness tests using alternative portfolio construction methodologies yield consistent results, with fourteen of twenty-five gross alpha specifications producing t -statistics

exceeding two. The strategy’s performance proves remarkably stable across different weighting schemes and breakpoint definitions, with value-weighted portfolios using market capitalization breaks generating a monthly alpha of 46 basis points (t-statistic = 4.42).

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CCC trading strategy maintains an alpha of 37 basis points per month with a t-statistic of 3.88. This result confirms that Cash Conversion Cushion captures a distinct dimension of production-based risk not subsumed by existing factors. The strategy’s factor loadings reveal a positive exposure to market risk (beta = 0.18, t-statistic = 7.43) and negative loadings on profitability and momentum factors, consistent with firms having different operational leverage and adjustment cost structures in production economies.

Our study extends the production-based asset pricing literature by identifying Cash Conversion Cushion as a novel proxy for firms’ exposure to systematic production risks, complementing existing work on profitability and investment factors [Fama and French \(2015\)](#); [Hou et al. \(2015\)](#). While prior research documents the pricing implications of accounting-based profitability measures, we demonstrate that cash conversion efficiency captures a distinct dimension of operational risk linked to marginal production costs and adjustment frictions. Unlike the gross profitability premium documented by [Novy-Marx \(2013\)](#), which focuses on revenues relative to assets, our measure emphasizes the quality of earnings through their convertibility to cash flows, providing a more direct link to production-based theories of asset pricing.

Methodologically, we contribute to the anomalies literature by conducting comprehensive robustness tests following the protocol of [Novy-Marx and Velikov \(2023\)](#), demonstrating that the Cash Conversion Cushion effect survives stringent controls for transaction costs and related factors. Our finding that CCC maintains significant alpha after controlling for 156 existing anomalies distinguishes it from many recently

proposed predictors that merely repackage known effects. The strategy’s performance in large-cap stocks, where market frictions are minimal, supports a risk-based interpretation consistent with production-based asset pricing models rather than behavioral explanations relying on limits to arbitrage.

Our results have broader implications for understanding the sources of cross-sectional return variation in production economies. The Cash Conversion Cushion’s predictive power highlights the importance of operational efficiency and working capital management as determinants of systematic risk exposure, suggesting that investors price firms’ ability to navigate production frictions and adjustment costs. This finding informs portfolio construction strategies by identifying a characteristic that captures exposure to production-based risks not spanned by existing factors, thereby expanding the efficient frontier available to investors. The economic magnitude and statistical robustness of the effect across different market segments and time periods establish cash conversion efficiency as a fundamental driver of expected returns in rational, production-based asset pricing frameworks.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in measurement across firms and to capture the most complete financial information available at fiscal year-end. Cash Conversion Cushion signal relies on two key financial statement variables. Cash (CH) represents the firm’s cash and cash equivalents as reported on the balance sheet, measuring the most liquid assets readily available to meet obligations and fund operations. EBITDA (EBITDA) captures earnings be-

fore interest, taxes, depreciation, and amortization, providing a measure of operating performance that excludes the effects of financing decisions, tax environments, and non-cash accounting charges. This metric represents the cash-generating capacity of the firm’s core operations before capital structure and tax considerations. We construct the Cash Conversion Cushion signal by dividing cash and cash equivalents (CH) by EBITDA (EBITDA) for each firm-year observation. This ratio captures the relationship between a firm’s liquid asset position and its operating cash generation capacity, providing insight into how many periods of current operating performance the firm’s cash holdings represent. We calculate this measure using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. For inclusion in our sample, we require non-missing values for both CH and EBITDA, and we exclude observations where EBITDA equals zero to avoid undefined ratios. Firms with negative EBITDA are retained in the sample, as negative values provide economically meaningful information about the relationship between cash holdings and operating performance during periods of operating losses.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CCC signal. Panel A plots the time-series of the mean, median, and interquartile range for CCC. On average, the cross-sectional mean (median) CCC is 0.42 (0.29) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CCC data. The signal’s interquartile range spans -0.96 to 1.43. Panel B of Figure 1 plots the time-series of the coverage of the CCC signal for the CRSP universe. On average, the CCC signal is available for 6.58% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does CCC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CCC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CCC portfolio and sells the low CCC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CCC strategy earns an average return of 0.34% per month with a t-statistic of 3.10. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.18% to 0.37% per month and have t-statistics exceeding 1.79 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 7.43 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 462 stocks and an average market capitalization of at least \$1,310 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 34 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for fourteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CCC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in nine cases.

Table 3 provides direct tests for the role size plays in the CCC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CCC, as well as average returns and alphas for long/short trading CCC strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CCC strategy achieves an average return of 28 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the CCC strategy relative to the five most common factor models range from 12 to 41 bps/month with t-statistics between 0.99 and 3.30.

5 How does CCC perform relative to the zoo?

Figure 2 puts the performance of CCC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CCC strategy falls in the distribution. The CCC strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CCC strategy (red line).² Ignoring trading costs, a \$1 invested in the CCC strategy would have yielded \$6.03 which ranks the CCC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CCC strategy would have yielded \$4.85 which ranks the CCC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the CCC relative to those. Panel A shows that the CCC strategy gross alphas fall between the 45 and 85 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CCC strategy has a positive net generalized alpha for five out of the five factor models. In these cases CCC ranks between the 62 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CCC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of CCC with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CCC or at least to weaken the power CCC has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CCC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CCC signal in these Fama-MacBeth regressions exceed -0.86, with the minimum t-statistic occurring when controlling for Idiosyncratic risk (AHT). Controlling for all six closely related anomalies, the t-statistic on CCC is 3.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the CCC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CCC strategy earns alphas that range from 33-37bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 3.22, which is achieved when controlling for Idiosyncratic risk (AHT). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CCC trading strategy achieves an alpha of 37bps/month with a t-statistic of 3.88.

7 Does CCC add relative to the whole zoo?

Finally, we can ask how much adding CCC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CCC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes CCC grows to \$2287.89.

8 Conclusion

This study provides robust evidence that the Cash Conversion Cushion (CCC) signal represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CCC is available.

Novy-Marx and Velikov (2023), demonstrates that CCC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CCC delivers an annualized Sharpe ratio of 0.41 gross (0.38 net of transaction costs), indicating strong performance that remains economically viable after implementation costs. Most notably, the strategy generates significant alpha of 37 basis points per month (t-statistic = 3.56) relative to the Fama-French five-factor model augmented with momentum, suggesting that CCC captures unique information about future returns not explained by conventional risk factors. The robustness of these results is further confirmed when controlling for six closely related strategies from the factor zoo, with the alpha remaining at 37 basis points per month (t-statistic = 3.88), demonstrating that CCC provides incremental predictive power beyond existing anomalies.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The persistence of CCC's predictive power after transaction costs suggests it could be profitably implemented in real-world portfolio strategies. The signal appears to capture fundamental information about firms' cash management efficiency that is not fully incorporated into market prices, potentially reflecting market inefficiencies or compensation for an unidentified risk factor.

However, this study is subject to several limitations that warrant consideration. First, while we employ conservative transaction cost estimates, actual implementation costs may vary depending on market conditions and trading infrastructure. Second, the analysis is limited to U.S. equity markets, and the generalizability of these findings to international markets remains unexplored. Third, the economic mechanisms underlying the CCC anomaly require further investigation to determine

whether the returns represent compensation for systematic risk or market mispricing.

Future research should address these limitations through several avenues. International studies could examine whether CCC's predictive power extends to global equity markets, providing insights into the universality of this phenomenon. Time-series analysis of the strategy's performance across different market regimes would enhance understanding of its risk characteristics and potential drawdowns. Additionally, investigating the microstructure and behavioral foundations of the CCC anomaly could illuminate why this signal generates persistent abnormal returns. Finally, examining the interaction between CCC and corporate financial policies, such as capital structure decisions and working capital management, may reveal the fundamental economic channels through which this signal operates. Such research would not only advance our theoretical understanding of asset pricing but also inform more sophisticated trading strategies that exploit inefficiencies in how markets process information about firms' cash conversion capabilities.

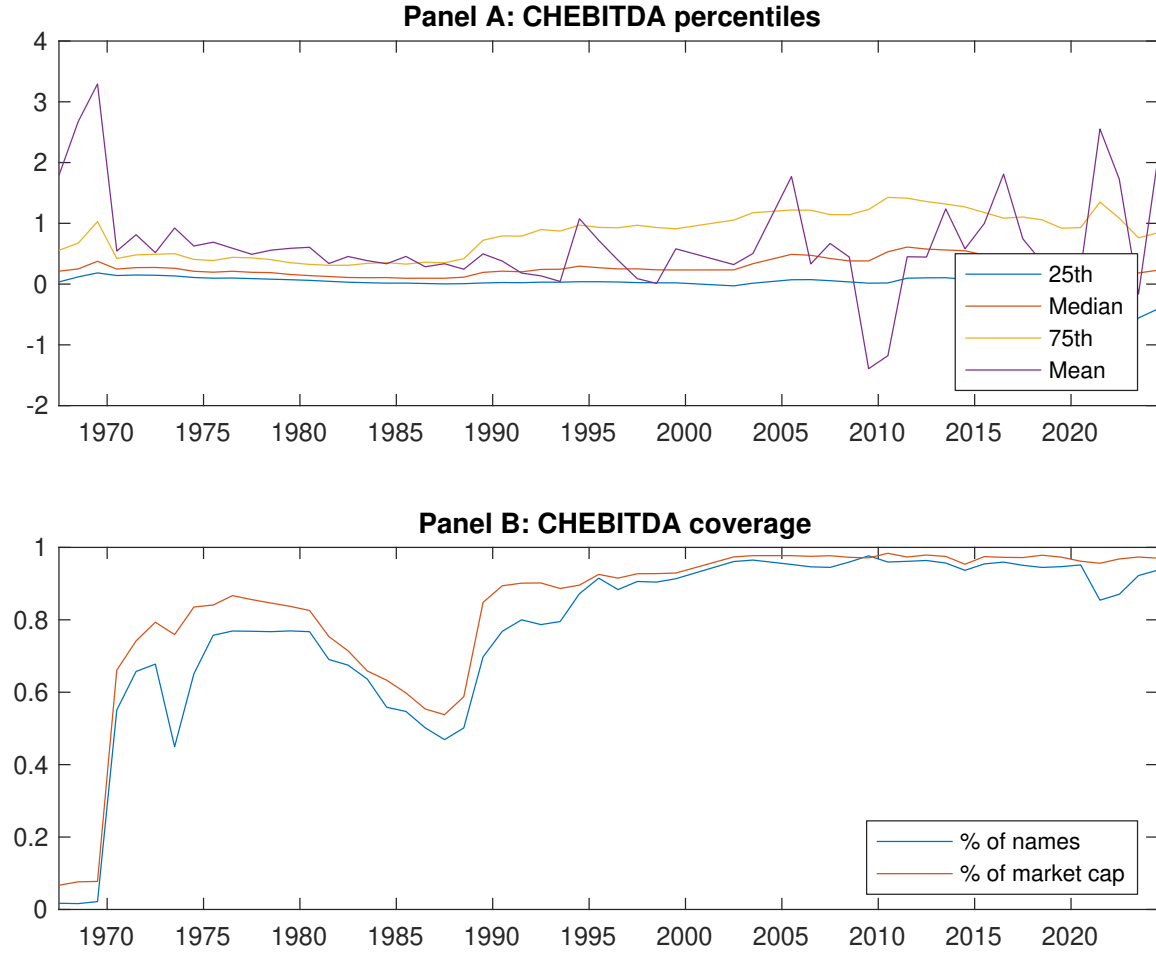


Figure 1: Times series of CCC percentiles and coverage. This figure plots descriptive statistics for CCC. Panel A shows cross-sectional percentiles of CCC over the sample. Panel B plots the monthly coverage of CCC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CCC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CCC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.59]	0.43 [2.37]	0.55 [2.91]	0.53 [2.57]	0.79 [3.68]	0.34 [3.10]
α_{CAPM}	-0.09 [-1.46]	-0.12 [-1.58]	-0.04 [-0.61]	-0.11 [-1.53]	0.11 [1.53]	0.20 [1.97]
α_{FF3}	-0.08 [-1.29]	-0.14 [-1.91]	-0.07 [-1.03]	-0.11 [-1.54]	0.10 [1.45]	0.18 [1.79]
α_{FF4}	-0.11 [-1.75]	-0.13 [-1.73]	-0.10 [-1.37]	-0.09 [-1.16]	0.15 [2.12]	0.26 [2.54]
α_{FF5}	-0.05 [-0.84]	-0.28 [-3.89]	-0.16 [-2.28]	-0.15 [-1.94]	0.26 [3.79]	0.31 [3.00]
α_{FF6}	-0.08 [-1.26]	-0.26 [-3.60]	-0.17 [-2.46]	-0.12 [-1.60]	0.29 [4.21]	0.37 [3.56]
Panel B: Fama and French (2018) 6-factor model loadings for CCC-sorted portfolios						
β_{MKT}	0.92 [58.96]	1.00 [57.78]	1.05 [63.13]	1.08 [59.41]	1.10 [68.09]	0.18 [7.43]
β_{SMB}	0.02 [0.77]	-0.05 [-1.88]	0.06 [2.61]	0.11 [4.25]	0.08 [3.22]	0.06 [1.62]
β_{HML}	0.02 [0.60]	0.01 [0.17]	0.04 [1.19]	-0.04 [-1.07]	0.07 [2.20]	0.05 [1.06]
β_{RMW}	-0.05 [-1.56]	0.34 [10.14]	0.19 [5.97]	0.11 [3.01]	-0.33 [-10.40]	-0.28 [-5.84]
β_{CMA}	-0.08 [-1.69]	0.10 [2.10]	0.08 [1.70]	0.02 [0.30]	-0.15 [-3.22]	-0.07 [-1.04]
β_{UMD}	0.04 [2.75]	-0.03 [-1.56]	0.02 [1.36]	-0.04 [-2.04]	-0.05 [-2.86]	-0.09 [-3.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1011	462	524	636	1003	
me (\$10 ⁶)	1310	2472	2653	2310	1846	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CCC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.10]	0.20 [1.97]	0.18 [1.79]	0.26 [2.54]	0.31 [3.00]	0.37 [3.56]
Quintile	NYSE	EW	0.36 [2.77]	0.44 [3.37]	0.34 [2.81]	0.26 [2.13]	0.12 [1.05]	0.07 [0.58]
Quintile	Name	VW	0.37 [2.20]	0.43 [2.60]	0.26 [1.74]	0.29 [1.88]	-0.06 [-0.40]	-0.01 [-0.10]
Quintile	Cap	VW	0.38 [3.42]	0.24 [2.33]	0.27 [2.58]	0.32 [3.08]	0.43 [4.13]	0.46 [4.42]
Decile	NYSE	VW	0.42 [3.10]	0.36 [2.65]	0.26 [1.94]	0.33 [2.46]	0.20 [1.50]	0.27 [1.97]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.32 [2.86]	0.16 [1.54]	0.14 [1.39]	0.19 [1.85]	0.24 [2.29]	0.27 [2.68]
Quintile	NYSE	EW	0.17 [1.27]	0.24 [1.79]	0.14 [1.13]	0.10 [0.80]		
Quintile	Name	VW	0.32 [1.92]	0.37 [2.21]	0.20 [1.37]	0.22 [1.46]		
Quintile	Cap	VW	0.35 [3.15]	0.20 [1.91]	0.22 [2.12]	0.25 [2.44]	0.34 [3.32]	0.36 [3.54]
Decile	NYSE	VW	0.39 [2.84]	0.32 [2.34]	0.22 [1.68]	0.26 [1.99]	0.17 [1.22]	0.21 [1.59]

Table 3: Conditional sort on size and CCC

This table presents results for conditional double sorts on size and CCC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CCC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CCC and short stocks with low CCC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CCC Quintiles					CCC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.30 [0.89]	0.51 [1.81]	0.67 [2.59]	0.81 [3.27]	0.93 [3.42]	0.62 [3.09]	0.77 [3.87]	0.60 [3.19]	0.53 [2.82]	0.24 [1.35]	0.21 [1.17]
	(2)	0.47 [1.62]	0.73 [3.00]	0.70 [2.91]	0.74 [3.09]	0.68 [2.62]	0.33 [2.02]	0.42 [2.58]	0.26 [1.70]	0.25 [1.60]	0.07 [0.44]	0.06 [0.43]
	(3)	0.51 [2.07]	0.55 [2.39]	0.82 [3.57]	0.65 [2.63]	0.63 [2.75]	0.09 [0.70]	0.11 [0.86]	0.02 [0.15]	-0.05 [-0.41]	-0.11 [-0.86]	-0.16 [-1.25]
	(4)	0.46 [2.21]	0.51 [2.33]	0.50 [2.25]	0.78 [3.54]	0.85 [3.61]	0.39 [2.98]	0.30 [2.30]	0.31 [2.44]	0.32 [2.46]	0.42 [3.21]	0.42 [3.20]
	(5)	0.47 [2.80]	0.44 [2.53]	0.40 [2.11]	0.48 [2.40]	0.75 [3.48]	0.28 [2.12]	0.12 [0.99]	0.14 [1.09]	0.22 [1.73]	0.36 [2.88]	0.41 [3.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CCC Quintiles					CCC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	412	409	420	422	422	31	32	40	42	39	
	(2)	117	112	117	117	113	62	62	66	67	63	
	(3)	79	78	83	80	78	106	107	115	109	105	
	(4)	63	63	63	64	63	217	225	224	225	226	
(5)	56	56	56	56	56	1248	2023	1990	1710	1475		

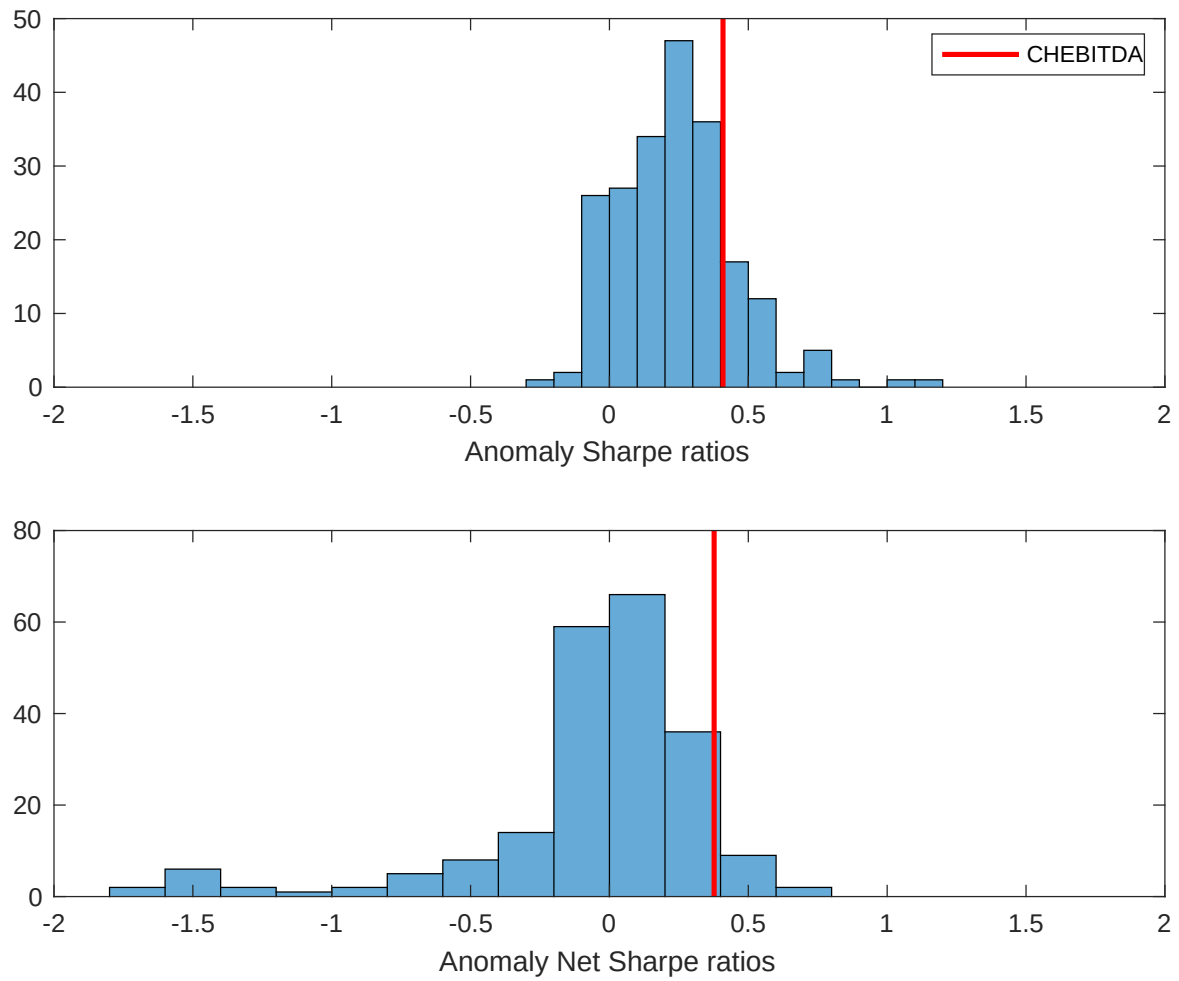


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CCC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

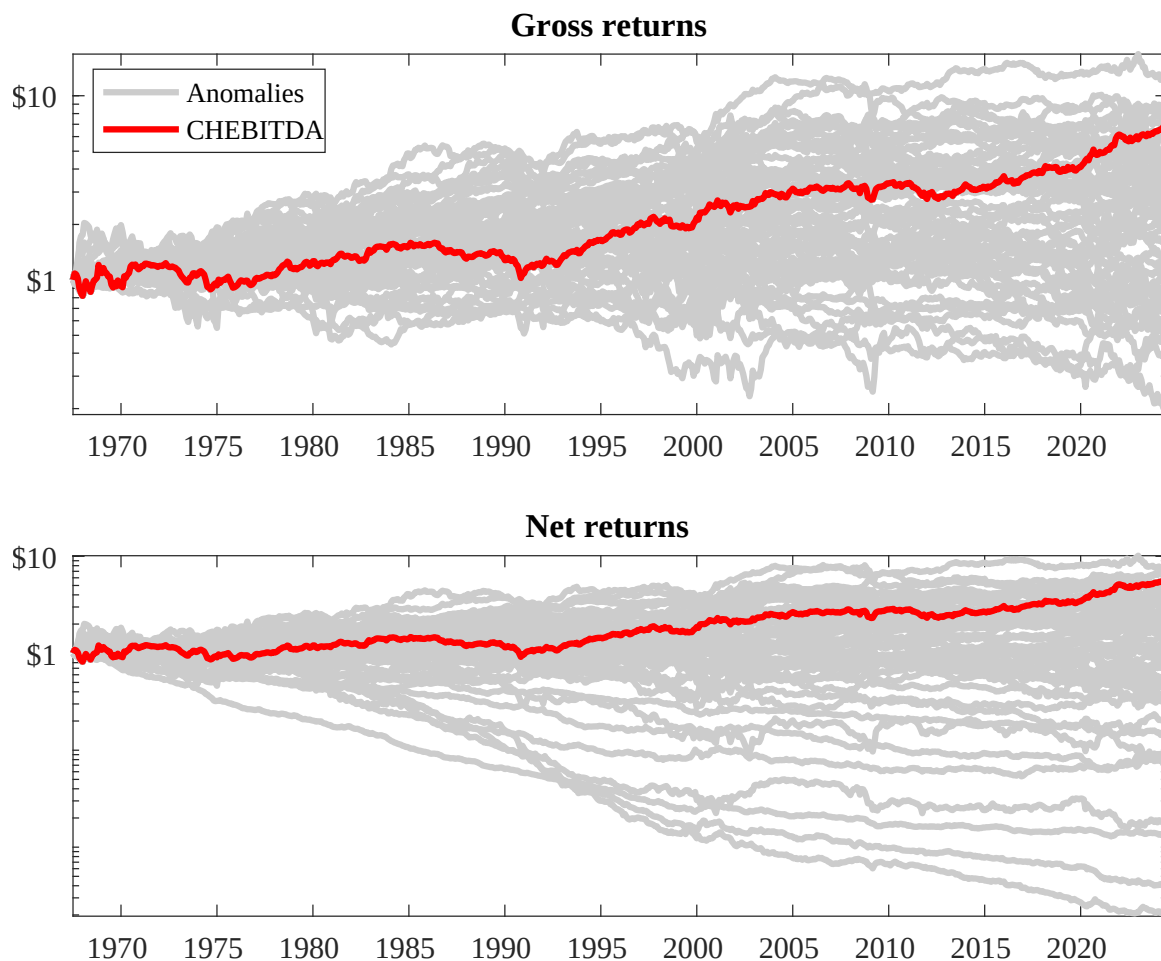
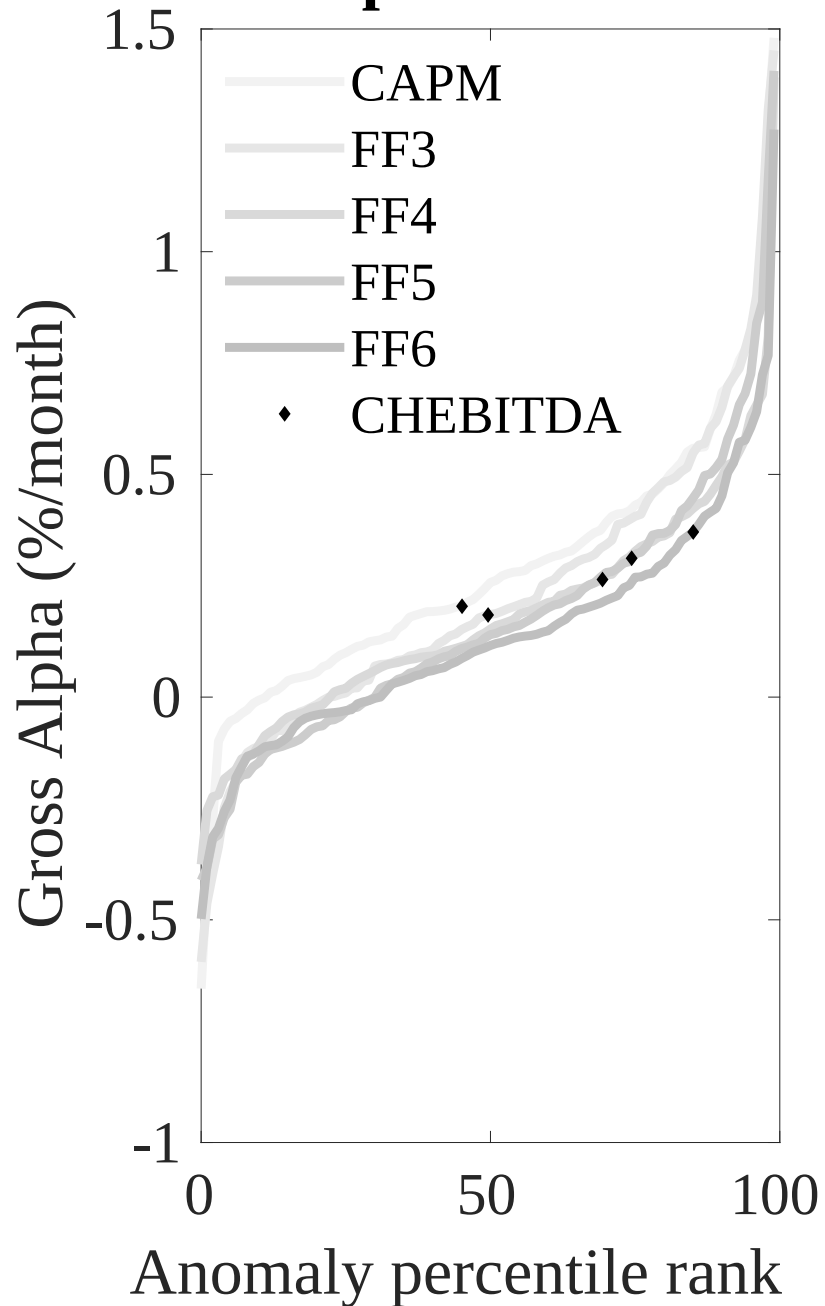


Figure 3: Dollar invested.

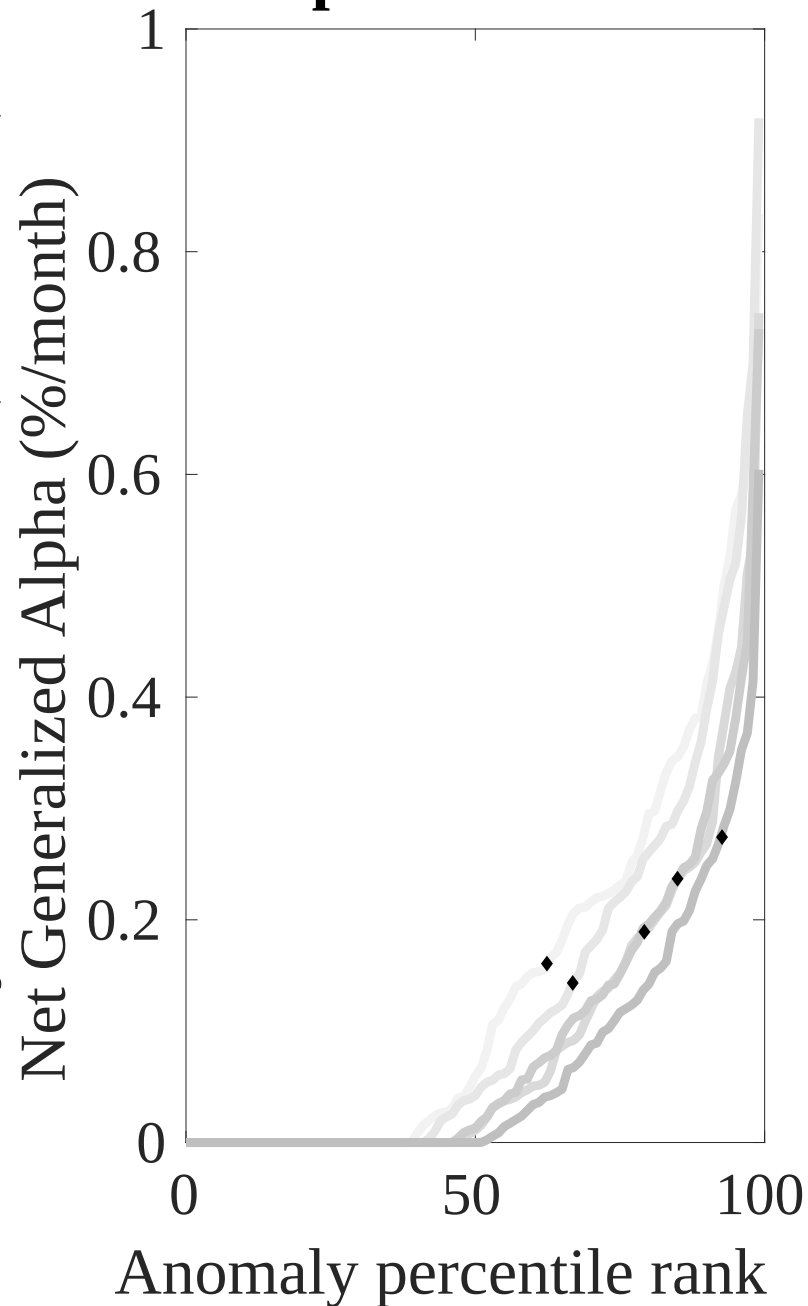
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CCC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



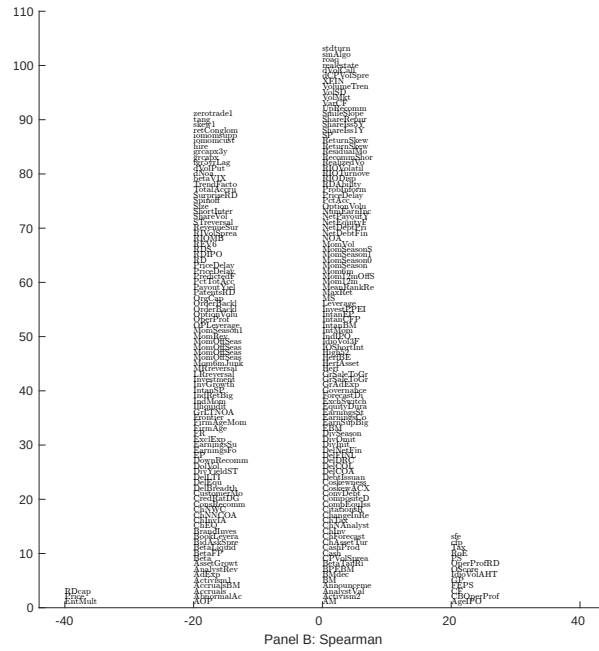
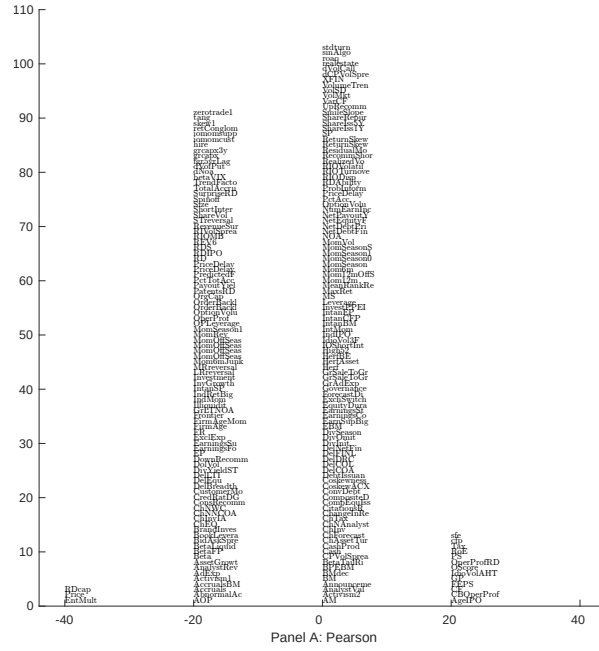


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CCC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

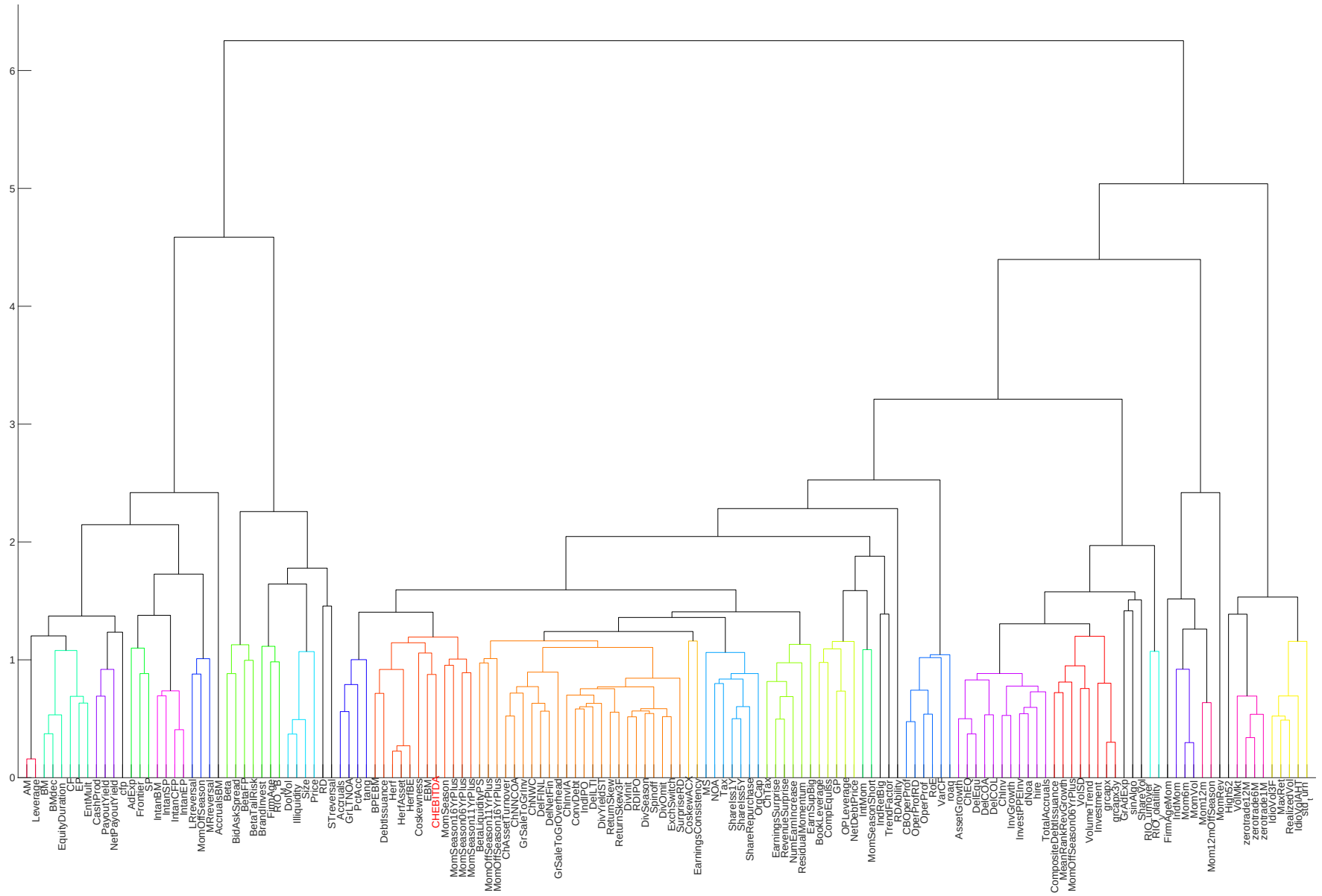


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

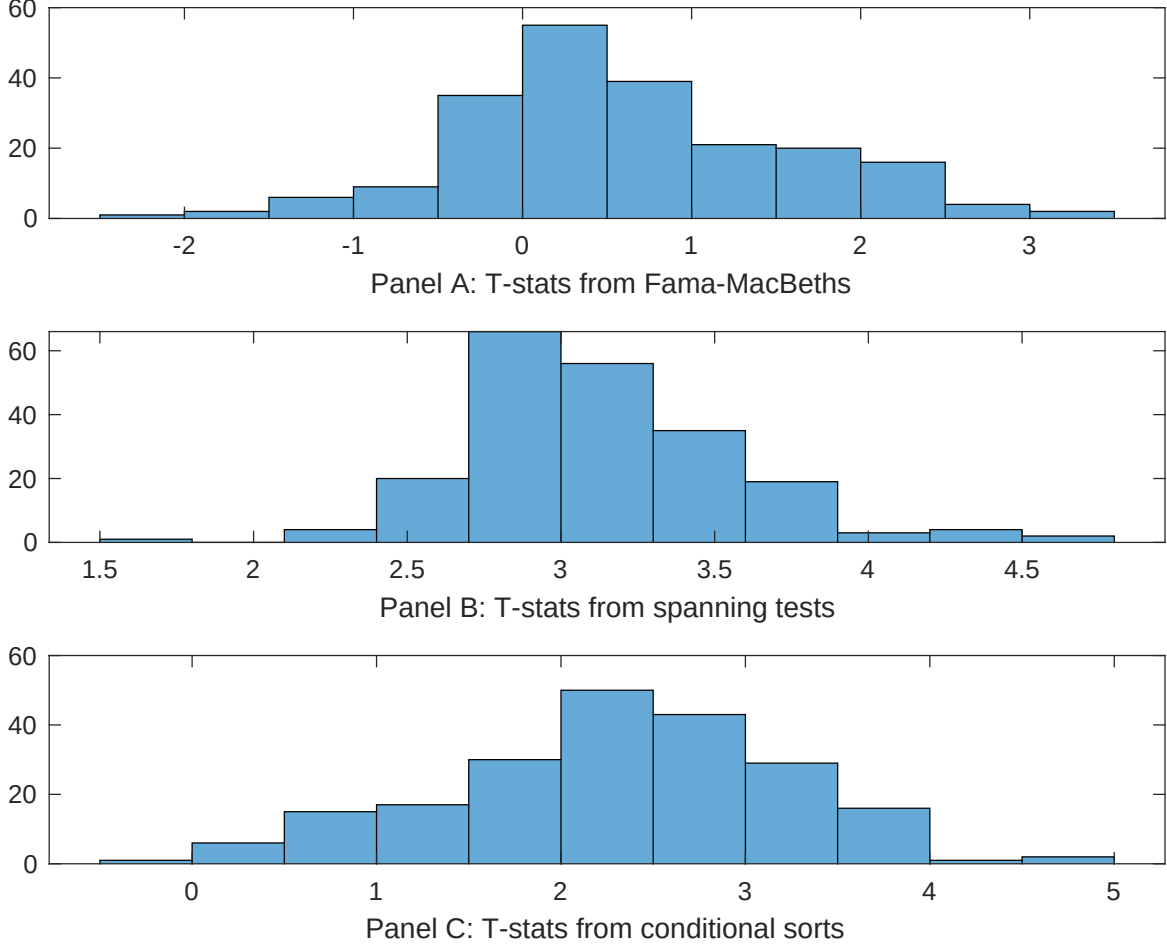


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CCC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.37]	0.14 [7.49]	0.12 [4.94]	0.13 [5.31]	0.15 [8.01]	0.12 [4.28]	0.16 [7.57]
CCC	0.58 [0.60]	-0.65 [-0.86]	-0.32 [-0.03]	0.22 [1.97]	-0.63 [-0.77]	0.14 [0.02]	0.55 [3.11]
Anomaly 1	0.84 [1.51]						0.12 [3.46]
Anomaly 2		0.95 [1.71]					-0.82 [-1.33]
Anomaly 3			0.47 [0.04]				0.10 [1.13]
Anomaly 4				0.19 [7.13]			0.12 [5.68]
Anomaly 5					0.16 [3.58]		0.27 [9.94]
Anomaly 6						0.61 [1.29]	0.17 [4.74]
# months	576	643	648	630	643	648	576
$\bar{R}^2(\%)$	2	2	0	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CCC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CCC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.37 [3.88]	0.34 [3.22]	0.36 [3.41]	0.33 [3.66]	0.34 [3.25]	0.36 [3.41]	0.37 [3.88]
Anomaly 1	4.75 [1.26]						7.58 [1.73]
Anomaly 2		1.01 [0.28]					-4.64 [-0.92]
Anomaly 3			-9.23 [-1.57]				-7.34 [-1.21]
Anomaly 4				2.46 [0.54]			9.37 [1.75]
Anomaly 5					-1.62 [-0.44]		-4.19 [-0.86]
Anomaly 6						-4.46 [-1.02]	-3.01 [-0.69]
mkt	18.46 [7.77]	18.85 [6.67]	17.01 [6.54]	17.51 [7.88]	17.90 [6.47]	19.05 [7.49]	16.52 [6.32]
smb	12.88 [3.04]	7.78 [1.53]	3.48 [0.86]	12.44 [3.56]	5.41 [1.13]	11.17 [1.90]	10.68 [1.95]
hml	6.73 [1.54]	5.52 [1.13]	4.79 [0.99]	8.46 [2.07]	6.28 [1.28]	6.42 [1.33]	9.47 [2.00]
rmw	-22.49 [-3.90]	-28.97 [-4.93]	-19.53 [-2.70]	-20.57 [-4.15]	-26.84 [-4.86]	-30.47 [-5.68]	-18.37 [-2.59]
cma	-26.19 [-4.09]	-7.73 [-1.06]	-9.61 [-1.33]	-26.61 [-3.91]	-6.71 [-0.92]	-7.35 [-1.03]	-31.13 [-4.13]
umd	-6.87 [-2.94]	-8.79 [-3.49]	-8.47 [-3.46]	-6.49 [-3.09]	-8.32 [-3.27]	-11.25 [-3.22]	-7.61 [-2.37]
# months	577	680	684	630	680	684	577
$\bar{R}^2(\%)$	27	22	22	29	22	22	28

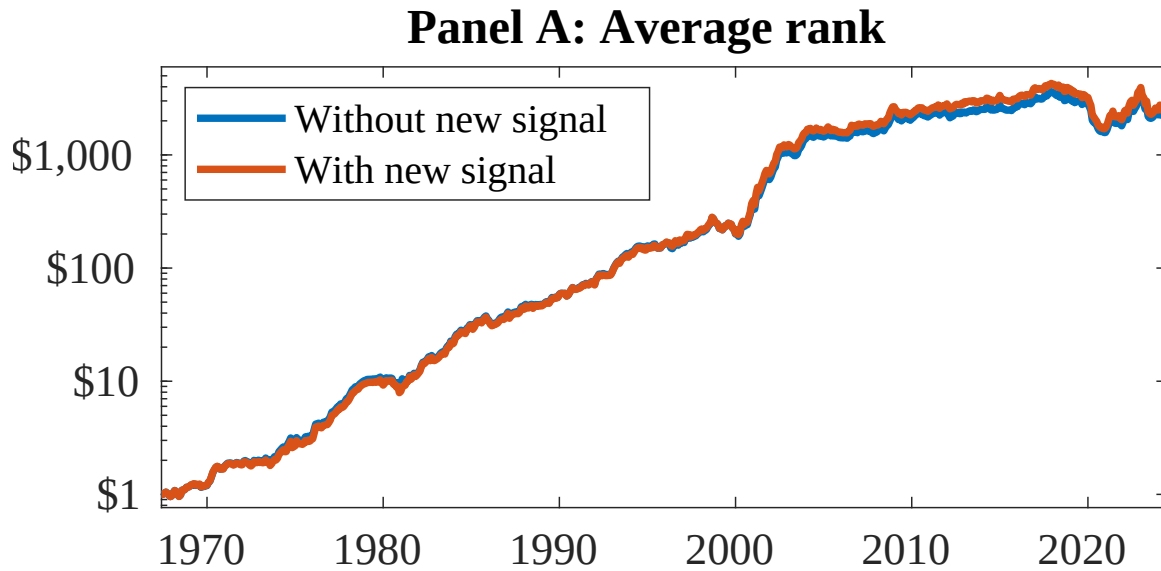


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CCC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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